

Architecture Audit Engine — Platform Overview

Executive feature brief: diagnostic ingestion, margin simulation, issue summaries & GraphQL remediation.

Executive Summary

The Shopify Revenue Assurance Audit Engine is a specialized diagnostic suite built to uncover silent margin erosion, zero-dollar cart vulnerabilities, unrestricted coupon stacking, and tracking pixel latency. It equips merchants, agencies, and auditors with instant risk assessment, interactive gross margin modeling, and executable GraphQL mutation playbooks.

01. Core Diagnostic Ingestion & Multi-Domain Evaluation

INGESTION & BENCHMARKS

- Direct Shopify Admin API & GraphQL configuration ingestion (discount nodes, delivery rules, web pixels).
- Pre-configured benchmark cases (e.g. Apex Performance Athletics, BFCM stacking, tracking latency).
- Discounts & Promotions Audit: CombinesWith flag inspection, runaway percentages, missing thresholds.
- Delivery & Shipping Audit: Post-discount subtotal free freight triggers, zero-dollar checkout loopholes.
- Tracking & Script Pixels: Redundant tags, unconsented scripts, and checkout friction detection.

02. Executive Risk Score & Margin Simulator

ANALYTICS & SIMULATION

- Store Risk Index (0–100 Dial): Weighted risk assessment (Critical, High, Moderate, Low Risk).
- Worst-Case Margin Exposure Calculator: Identifies maximum possible stackable discounts & \$0 cart exploits.
- Interactive Order Margin Simulator: Sandbox modeling order subtotals, COGS, shipping expenses, and stacked coupon codes with real-time gross/net profit visualization.

03. Detailed Audit Findings & Raw JSON Export

INSPECTION & EXPORT

- Interactive filtering by severity (Critical, High, Medium, Low) and operational domain.
- Financial exposure statements detailing dollar impact and margin leakage mechanisms.
- Direct Shopify Admin navigation click-paths (e.g. Discounts > Automatic > Combination Rules).
- Full JSON audit report viewer with syntax formatting, copy to clipboard, and local file download.

04. Operational Remediation Playbook & Status Tracker

GRAPHQL AUTOMATION

- Production-ready GraphQL Admin API mutation snippets ready for GraphiQL and custom apps.
- Interactive issue status tracker: Flag findings as Open, In Progress, or Resolved with custom auditor notes.
- Bulk script export: "Copy All Mutations" button aggregates executable remediation steps into one script.

05. Email Issues Summary & Extensible Account Service

EXECUTIVE REPORTING

- Executive Issues Recap: Highlights Store Risk Index, \$0 cart exploit exposure, and severity distributions.
- Itemized vulnerability overview with fix recommendations and custom auditor commentary.
- Extensible recipient architecture: Decoupled for future Shopify App Bridge / OAuth / Firebase tokens.
- Dedicated backend API (/api/send-recap) with transmission logging and archive audit log (/api/recaps).

06. Developer Experience, Security & Usability

INFRASTRUCTURE

- GraphQL Query Helper Modal: Built-in master query for extracting full store metadata via Shopify API.
- Auditor Feedback Modal with backend rate-limiting (1 submission per 5 mins per IP) and countdown timer.
- Dark, Light, and System theme synchronization with responsive, single-screen dashboard layout.

Technical Stack & Architecture Specifications

- Framework: React 19 + TypeScript + Vite | Styling: Tailwind CSS | Icons: Lucide Icons
- Backend: Express server with secure /api endpoints | API Compatibility: Shopify Admin API 2024-10+
- Client-Side Integrity: Pure memory state engine with zero unprompted data persistence.